

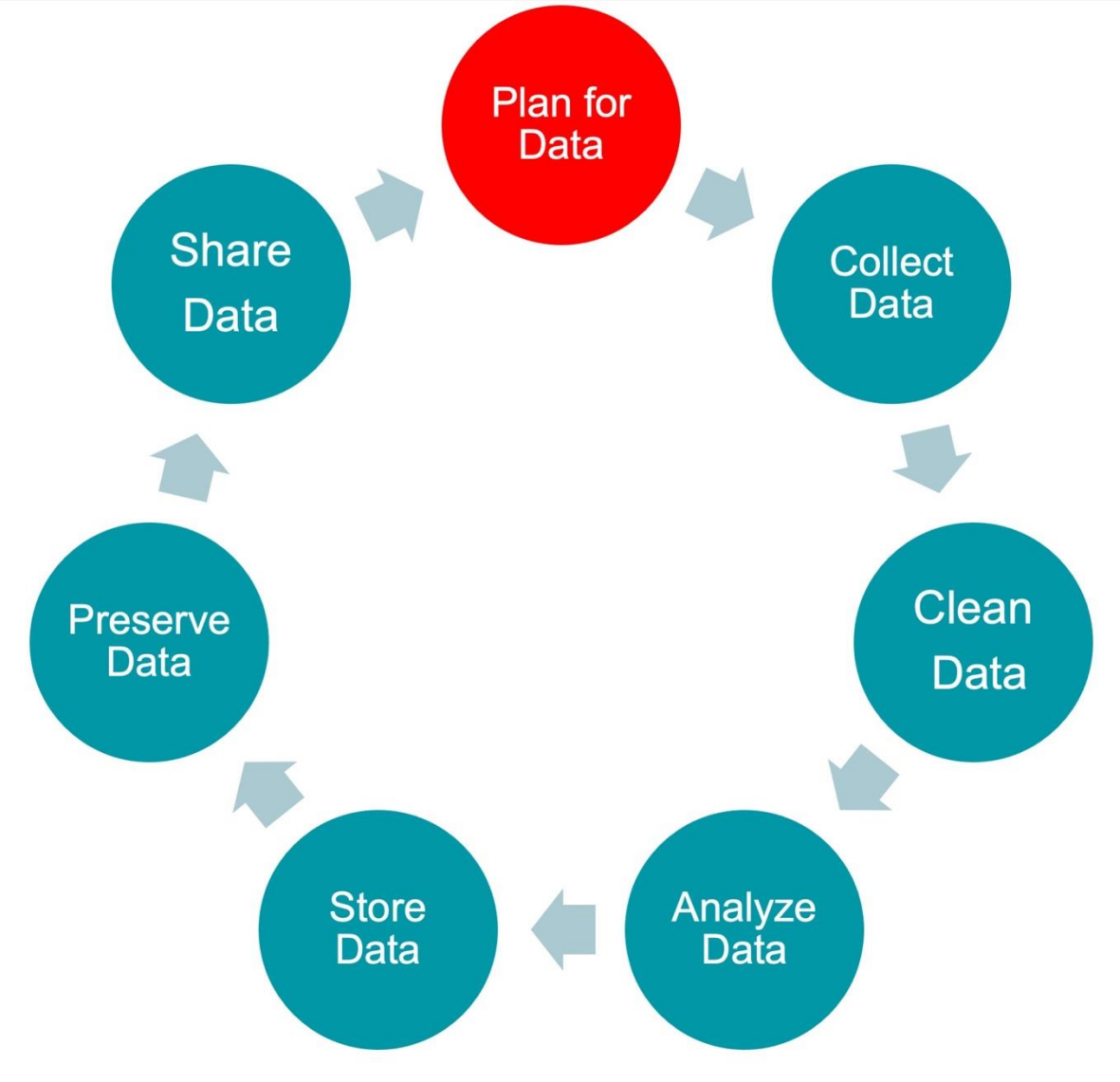
Royal Netherlands Institute for Sea Research

FAIR - UNECSS workshop

Best Practices for Making (Meta)Data FAIR

Antsje van der Leij & Flavio Francisco

Alkmaar - Netherlands



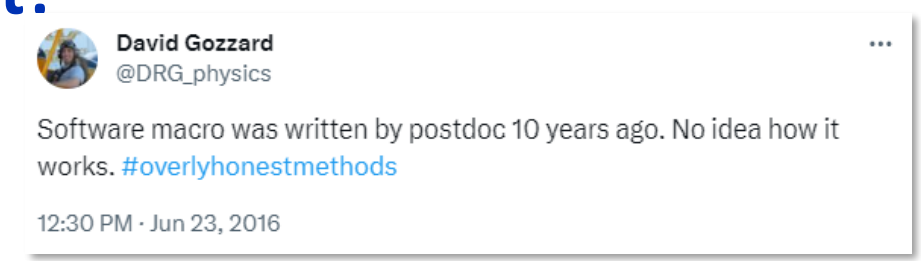
Outline

- **Why** should we plan for data management?
- **What** is covered in a data management plan (DMP)?
- **Who** – shared responsibility
- **When –** living document
- Tips
- Resources
- Wrap up

Why should we plan for data management?

What could go wrong?

- 3 years into the project, data is lost
- Data exists but can't be reused (no documentation, corrupted, outdated format…)
- You are required to share data but you didn't ask for consent
- Disagreement between project partners on exploitation of results
- Can't keep track of your own files





What is included in a DMP?

- **Project description** – brief: setting the scene, *not* explaining the science
- **Description of data** – types, sizes, origin; raw, processed, and final data
- **Responsibilities** – who is responsible for the data during the project? Who (person or organisation) is responsible for the data after the project has ended?
- **Data storage** – where will data be kept during the project. Are backups done?
- **Data longevity and security** – where will data be stored for the long-term (e.g. by the repository where you publish it, or in your employer's / university's long-term storage?)
- **Ethics** – are there any ethical considerations to take into account regarding your collected data? Do you have to consider GDPR (personal information about people?)

What is included in a DMP?

Resources:
Horizon Europe DMP template: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/temp-form/report/data-management-plan_he_en.docx

ARGOS is a platform for creating structured, shareable plans that define how data, software, and other digital outputs are managed. https://argos.openaire.eu/portal/

Create a Data Management Plan Using the AQUASERV Blueprint. https://www.aquaserv-ri.eu/ta-handbook

Example DMP: AQUASERV DMP : FishEdge https://zenodo.org/records/17272382

What is included in a DMP?

The FAIR data principes

"Check list"

FINDABLE	Persistent identifiers (PIDs)	Rich metadata with core elements	Metadata includes PID	Machine readable metadata
ACCESSIBLE	Metadata has access level & conditions	Standard communication protocol	Metadata is always available	
INTEROPERABLE	Metadata in formal KR language	Vocabularies & ontologies	Linked metadata	
REUSABLE	Metadata specifies data content	Metadata includes license	Metadata includes provenance	(Meta)data community standards

What is included in a DMP?



Findable:

- ❑ **F1, F3, F4** List those repositories via which you will share your data
  - Can they give you dataset DOIs?
  - Do they provide for good metadata?
- ❑ **F4** Are there data you will not publish.
  - Why?
  - Will you make a metadata record for those data, even if you do not attach the data to it?
  - How will access to interested parties be made possible?
- ❑ **F2** How and what metadata will you collect during your project?
  - Will those be included with your data package
  - Will those be included via the repository's own metadata?

What is included in a DMP?



Accessible:

- ❑ **A1** What does the repository offer you for accessiblity?
  - Does the dataset and the metadata record have a DOI or URI?
  - For how long will the repository keep the record(s) alive?
- ❑ **A2** Bear in mind that a FAIR repository will never delete metadata records. The data may be, but the metadata record should always remain.
- ❑ Do **you** have additional access restrictions?
  - Embargo periods: who will then release the embargo, and when?
  - Sensitive data or privacy issues: who can then access those data?
- ❑ Do you foresee any **difficulties** with large data, proprietary formats, etc?

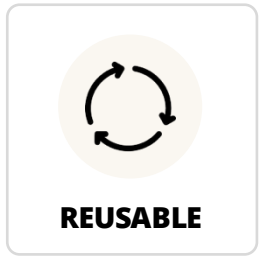
What is included in a DMP?



Interoperable:

- ❑ **I1** Standards!
  - What data standards and formats do you want to use?
  - What requirements do the chosen repositories make?
  - Are there some data you cannot make Interoperable? How then will someone open and use them?
- ❑ **I2** Vocabularies and formatting
  - Choose the vocabularies to use for keywords, measurements. Decide how they should be written into your data.
  - What number and date formatting to use?
  - Create data collecting templates so data are interoperable from the beginning
- ❑ **I3** Document your data trail -- if one data is based on another, then document that link. If you download data from somewhere, keep a copy of the URL/DOI and the metadata record

What is included in a DMP?



Reusable:

- ❑ **R1** Provenance, licences
  - What licences and additional access constraints? Why?
  - What provenance information will you collect? Make a template.
  - How will that provenance information be made available to data users?

Data repositories and FAIR

A shared responsibility

FAIR principle	In practice	Repository	Researcher
FINDABLE	Assign a persistent identifier (PID)	●	
	Data are described with rich metadata	●	●
	Metadata include the PID, are indexed & searchable online	●	
ACCESSIBLE	(Meta)data retrievable via PID using a standardized communications protocol (e.g. html)	●	
	Authentication and authorization procedure	●	
	Metadata remain accessible (even if data aren't)	●	
INTEROPERABLE	Metadata standards	●	
	Controlled vocabularies	●	●
	References to other (meta)data	●	●
REUSABLE	(Machine readable) license / terms of use	●	●
	Detailed provenance documentation		●
	Domain standards	●	●

The DMP Journey: FAIR by desing

Make an inventory of your research data/materials

Explore potential data repositories and their requirements & guidelines.
- Inc. use of standards, vocabularies, etc.

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

FAIRsharing

DCC

Adopt domain standards during data collection & processing

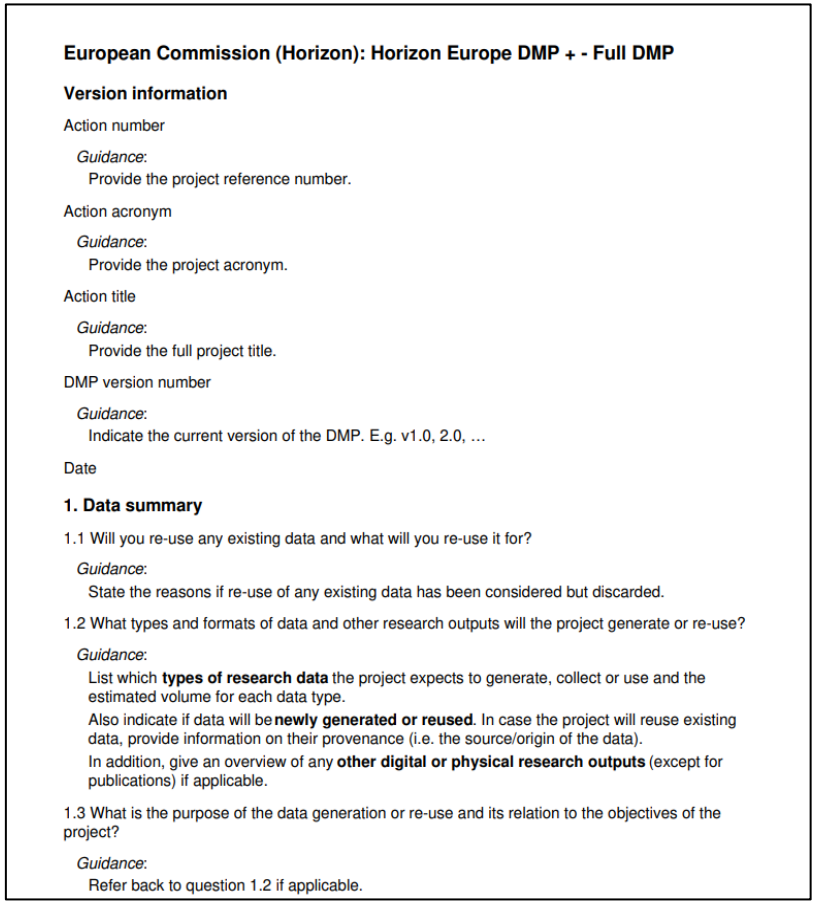
Document as you go
- Notebook
- Codebook, data dictionary
- Workflows
- Protocols
- Readme files
- Data overview with metadata
- File organization and naming

Deposit data & documentation in repository; describe it with rich metadata, inc. access & reuse conditions

General tips for DMP

- Be concrete
  - Specify management actions for specific datasets/materials
- Be coherent across sections
- Be honest
  - Don’t state things that you do not intend to do
- Be pro-active
  - It’s OK not to have all answers at the beginning of the project
  - Start with the end (publishing) in mind, explore repositories and their requirements early on

Look for a funder template!



European Commission (Horizon): Horizon Europe DMP + - Full DMP

Version information

Action number

Guidance:
Provide the project reference number.

Action acronym

Guidance:
Provide the project acronym.

Action title

Guidance:
Provide the full project title.

DMP version number

Guidance:
Indicate the current version of the DMP. E.g. v1.0, 2.0, …

Date

1. Data summary

1.1 Will you re-use any existing data and what will you re-use it for?

Guidance:
State the reasons if re-use of any existing data has been considered but discarded.

1.2 What types and formats of data and other research outputs will the project generate or re-use?

Guidance:
List which **types of research data** the project expects to generate, collect or use and the estimated volume for each data type.
Also indicate if data will be **newly generated or reused**. In case the project will reuse existing data, provide information on their provenance (i.e. the source/origin of the data).
In addition, give an overview of any **other digital or physical research outputs** (except for publications) if applicable.

1.3 What is the purpose of the data generation or re-use and its relation to the objectives of the project?

Guidance:
Refer back to question 1.2 if applicable.

DMPonline.be

Useful resources

- How to properly describe your research data
- DMP examples:
  - LIBER DMP Catalogue
  - Examples on the Digital Curation Centre (DCC) website
  - DMPs published in the *Research Ideas and Outcomes* open access journal
  - Public Horizon 2020 DMPs collected by OpenAIRE Austria
- Knowledge clip: the FAIR data principles
- Self-paced online course: Fundamentals of research data management
- Creating a DMP:
  - https://argos.openaire.eu/home
  - https://dmponline.be/

What is included?

Data summary & description

Purpose, origin, type, sensitiveness, format & volume

Ethical & legal issues

Intellectual property, GDPR, etc.

Roles & responsibilities

Data storage & security

Security measures, back-up, data transfers & access control

Documentation & metadata

Information to keep data findable, understandable, reusable

Costs & resources

Long-term preservation

What to preserve & where, for how long, access restrictions

Data sharing and reuse

Data repositories, access conditions, licenses or agreements



QUESTIONS AND HANDS ON

